

NIKHIL SAI TATA

AI / ML Engineer | LLMs, RAG & Generative AI | Machine Learning & Computer Vision

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SUMMARY

AI/ML Engineer and 2026 Computer Science graduate with hands-on experience building **AI agents**, **LLM-powered applications** and **RAG pipelines** on AWS. Built an NL-to-SQL AI agent on Amazon Bedrock with **95%+ query accuracy** and a computer-vision forgery detector with **95% accuracy**. Skilled in Python, PyTorch, Scikit-learn, LangChain and prompt engineering, with full-stack skills to deploy AI into production.

TECHNICAL SKILLS

Generative AI & LLMs: LLM APIs (OpenAI, Anthropic Claude, Groq), RAG, LangChain, Prompt Engineering, Embeddings, Ollama

Machine Learning: Scikit-learn, PyTorch, Hugging Face Transformers, Random Forest, SVM, NLP, Computer Vision, Feature Engineering

Data: Pandas, NumPy, Matplotlib, Data Preprocessing, Exploratory Data Analysis (EDA), SQL, MySQL

AWS & Cloud: Amazon Bedrock, Bedrock Agents & Knowledge Bases, Titan Embeddings, OpenSearch, Lambda, S3, RDS, ECS, IAM, CloudWatch

Languages: Python, JavaScript, SQL, C++

Development & Tools: Django, React.js, Node.js, REST APIs, Docker, Git, GitHub, Linux, Postman, n8n

Core CS: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks

EXPERIENCE

Coding and Programming Intern | Samsung Innovation Campus (NSIC)

Sep 2024 – Oct 2024

Project: AskSQL – Natural Language to SQL System (Generative AI)

- Architected **AskSQL**, an AI-powered NL-to-SQL system using **Amazon Bedrock and Anthropic Claude** that converts plain-English prompts into SQL, letting non-technical users query live business data without manual reports.
- Built a **RAG-based semantic retrieval pipeline** with Amazon Titan Text Embeddings v2 and Bedrock Knowledge Base, improving schema understanding and reducing query ambiguity across complex relational datasets.
- Integrated AWS Lambda and Amazon RDS to **automate end-to-end SQL generation**, execution and real-time response delivery.
- Achieved **95%+ query accuracy** across 100+ advanced aggregation queries through prompt engineering and few-shot learning.

Data Analyst Intern | Vodafone Idea (AICTE)

Oct 2024 – Nov 2024

- Cleaned and analyzed **2+ lakh structured records** in Python using preprocessing, validation and missing-value handling.
- Developed **Python data analysis and visualization workflows** to identify seasonal agricultural yield patterns and speed up decisions.
- Performed **EDA and statistical trend analysis** to uncover actionable insights for field-level research and operational reporting.
- Automated data cleaning and reporting workflows with **Pandas, NumPy and Matplotlib**, reducing manual effort.

PROJECTS

Automated Document Forgery Detection using Computer Vision & ML | Python, Scikit-learn, Django

GitHub

- Developed a **computer vision and machine learning system** that detects tampering in uploaded PDF and image documents.
- Engineered a **feature-extraction pipeline** using Error Level Analysis (ELA), image preprocessing and forensic analysis.
- Trained and compared classifiers, achieving **95% detection accuracy with Random Forest** and 92% with SVM.
- Built **model inference**, metadata inspection and report generation, with a web interface that highlights suspected tampered regions.

KnowYC – Verified Student Community Platform | React, Node.js, Express, MySQL

Live: knowyc.org

- Built and deployed a **full-stack platform** for college students with **149 REST APIs**, a 29-table MySQL schema, secure auth and RBAC.
- Built a document upload pipeline with **PDF/DOCX text extraction and rule-based relevance screening** plus admin moderation.
- Implemented **geospatial search** (MySQL ST_Distance_Sphere, Leaflet) and a 13,000+ college directory search returning results in ~30 ms.
- Deployed to production at **knowyc.org** on Render with a TLS-secured Aiven MySQL database at zero hosting cost.

LEADERSHIP & CERTIFICATIONS

Google Student Ambassador

Aug 2025 – Dec 2025

- Organized and delivered **4 hands-on Generative AI workshops for 500+ students**, driving campus adoption of Google Gemini through training in prompt engineering and practical AI tools.

Postman API Fundamentals Student Expert | Postman | [Verify Credential](#)

Oct 2025

EDUCATION

Bharath Institute of Higher Education and Research, Chennai, India

2022 – 2026

Bachelor of Technology in Computer Science

CGPA: 8.44 / 10

Narayana Junior College, Hyderabad | Intermediate (Class XII), MPC | **95.6%**

2020 – 2022

Ideal High School, Suryapet | SSC (Class X) | **CGPA: 10 / 10**

2019 – 2020